

Probability & Sample Space

WorksheetHub AI
Mathematics · Grade 7

Unit 4 · Data & Probability

Medium

Probability

Name: _____

Date: _____

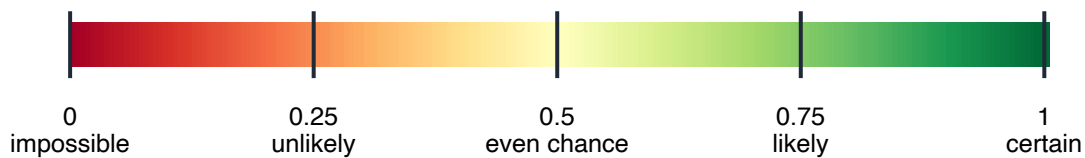
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Remember

Probability = $\frac{\text{number of ways the event can happen}}{\text{total number of equally likely outcomes}}$. It is always between 0 and 1.

Experimental probability = $\frac{\text{number of times it happened}}{\text{number of trials}}$. **Expected number** = probability $\times$ number of trials.

Probability scale



Worked example

Roll a fair dice. $P(\text{even}) = \frac{3}{6} = \frac{1}{2}$ because 2, 4, 6 are even out of 6 outcomes.

Sample space: two dice added together

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

36 equally likely outcomes. Use this table for questions 4–6.

A. Single events

- 1 A fair dice is rolled. Find $P(\text{a prime number})$.

Answer:

- 2 A bag has 3 red, 5 blue and 2 green counters. One is picked at random. Find $P(\text{blue})$.

Answer:

- 3 Using the spinner picture, find $P(\text{yellow})$ and $P(\text{red or blue})$.

Answer:

B. Two dice (use the sample space table)

- 4 How many outcomes give a total of 7? Find $P(\text{total } 7)$.

Answer:

- 5 Find $P(\text{total of } 10 \text{ or more})$.

Answer:

- 6 Find $P(\text{total of } 1)$.

Answer:

C. Experimental probability

- 7 A dice is rolled 50 times and a six comes up 8 times. Find the experimental probability of a six.

Answer:

- 8 A fair dice is rolled 120 times. How many sixes would you expect?

Answer:

- 9 A coin is flipped 40 times and lands on heads 26 times. Is the coin definitely unfair? Explain.

Answer:

D. Listing outcomes

- 10 A coin is flipped and a dice is rolled. List all the possible outcomes (e.g. H1). How many are there?

Answer:

- 11 From your list, find $P(\text{heads and an even number})$.

Answer:

★ Challenge

- 12 A bag has red and blue counters only. $P(\text{red}) = 0.4$ and there are 12 blue. How many counters are in the bag?

Answer:

- 13 Two fair dice are rolled. Which total is most likely and what is its probability?

Answer: